

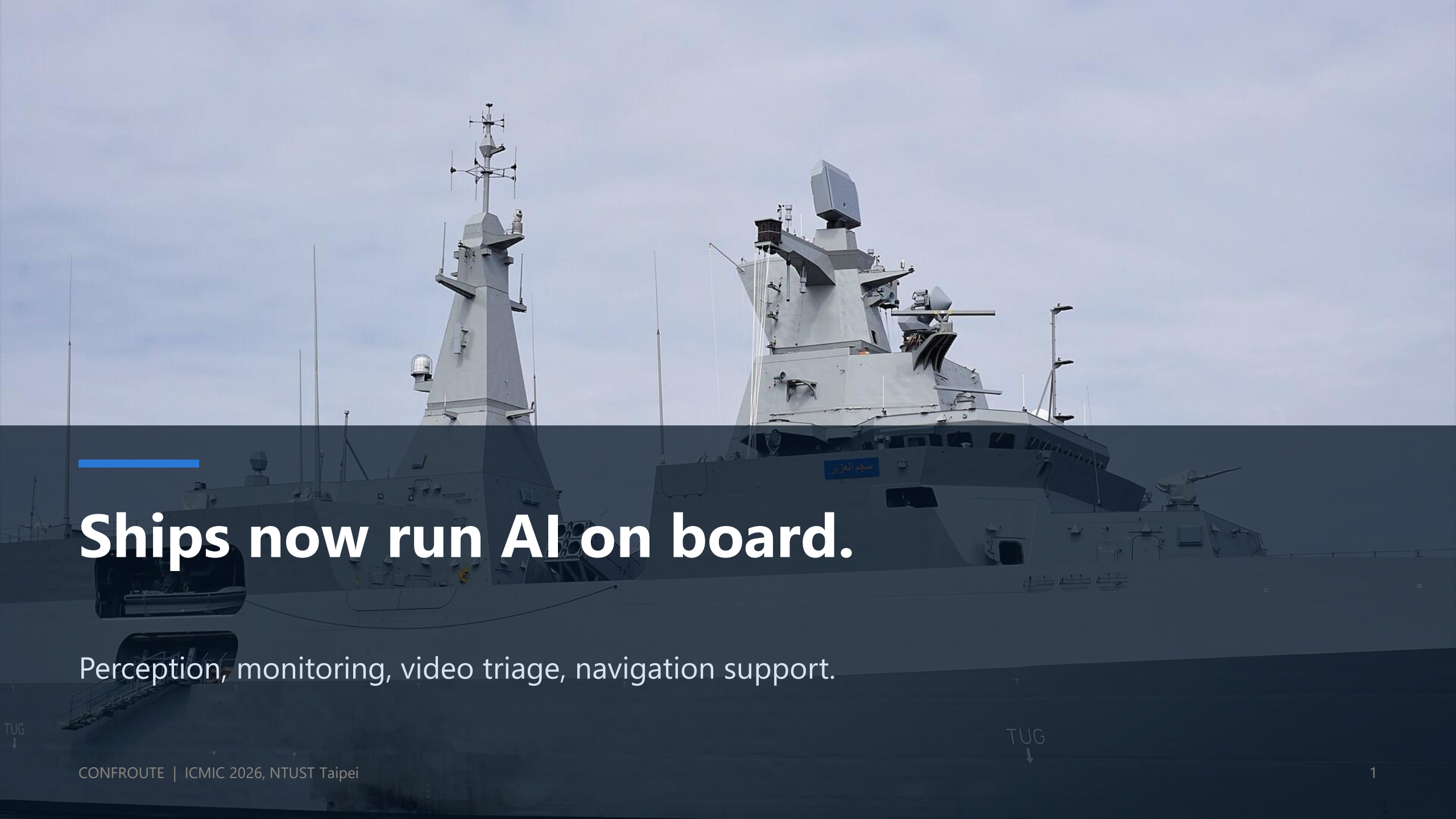
ICMIC 2026 · NTUST, Taipei · 19–21 August 2026

CONFROUTE

Confidence-Aware Routing for Resilient Maritime
Edge AI

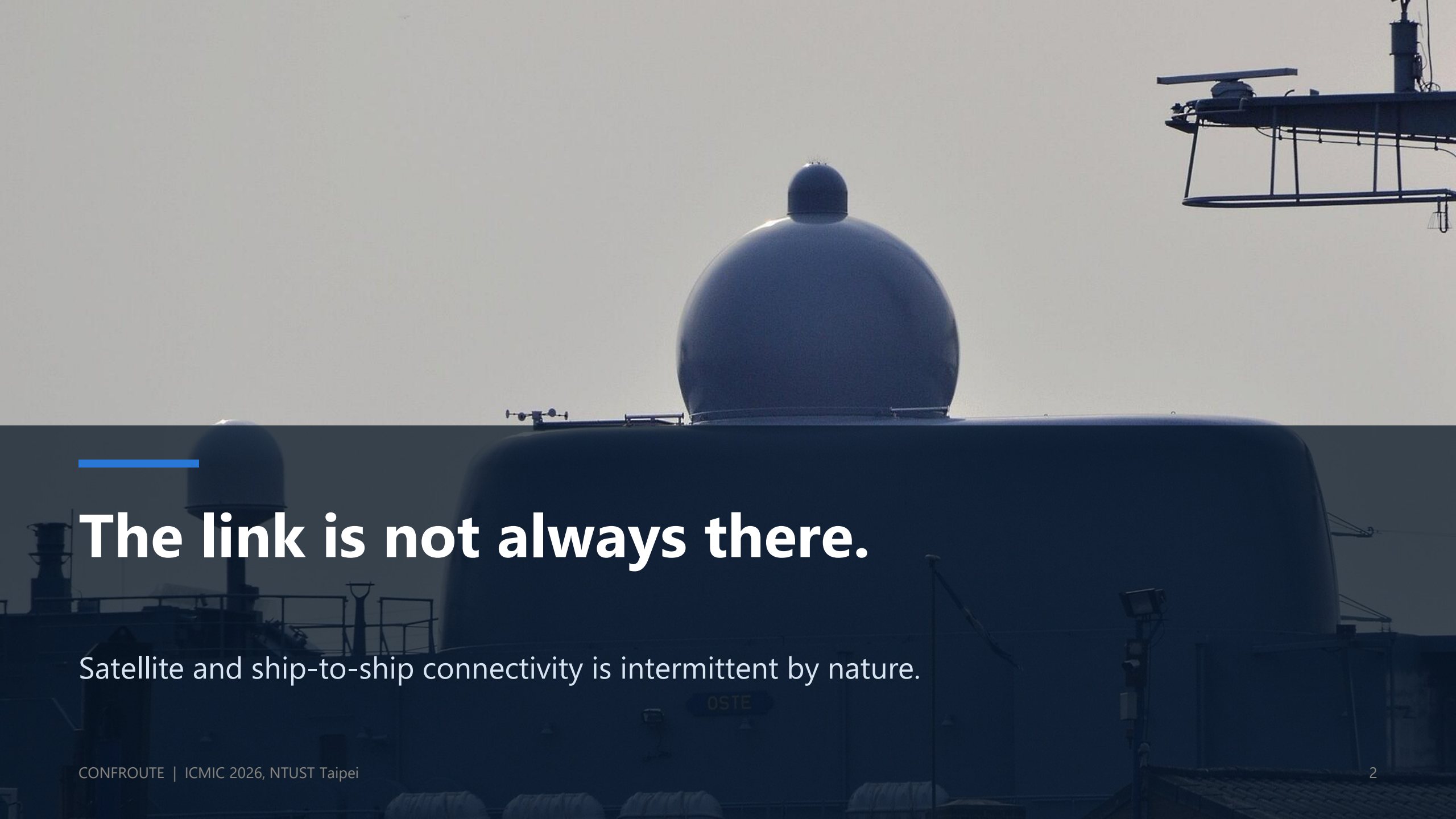
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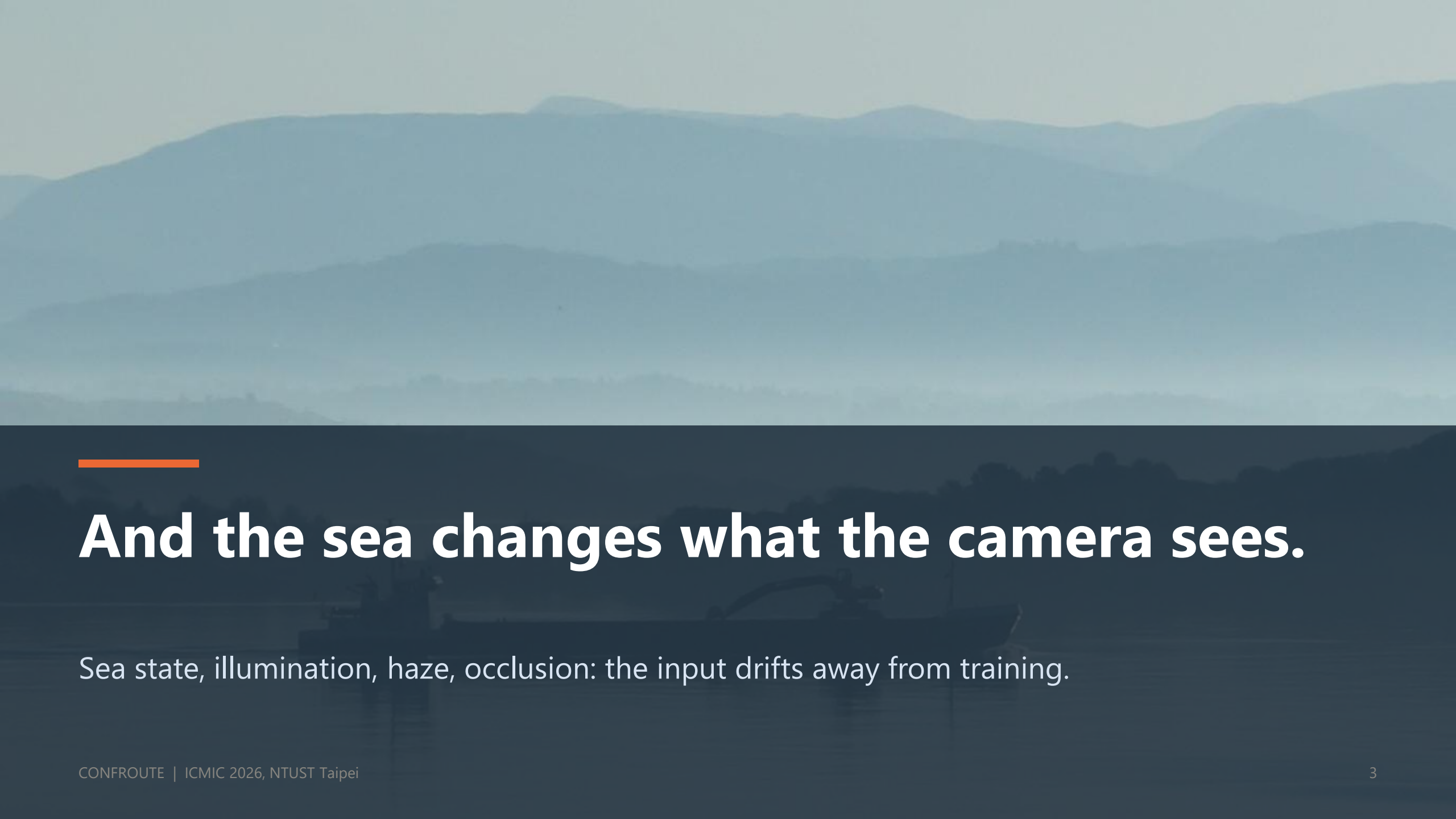
Ships now run AI on board.

Perception, monitoring, video triage, navigation support.



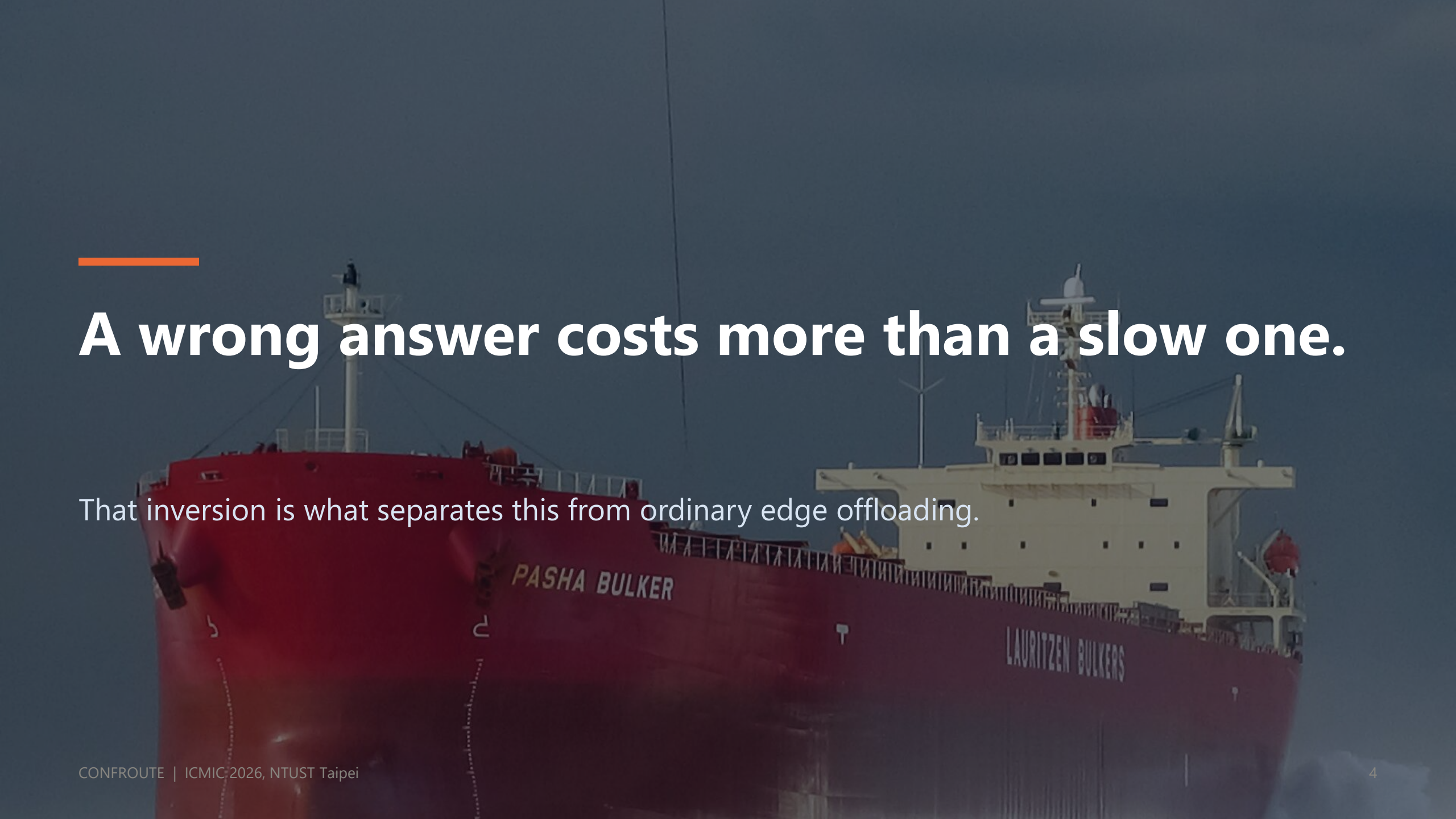
The link is not always there.

Satellite and ship-to-ship connectivity is intermittent by nature.



And the sea changes what the camera sees.

Sea state, illumination, haze, occlusion: the input drifts away from training.



A wrong answer costs more than a slow one.

That inversion is what separates this from ordinary edge offloading.

Act now, ask for help, or refuse to answer?

Every task, on every node, under uncertainty.

Everyone assumes the model knows when it is unsure.

Confidence-threshold offloading is built entirely on that assumption.

Clear water: the classifier finds every ship.

Three held-out ShipsNet chips, 80 × 80 PlanetScope imagery.

Nominal



"SHIP" conf 0.67
correct

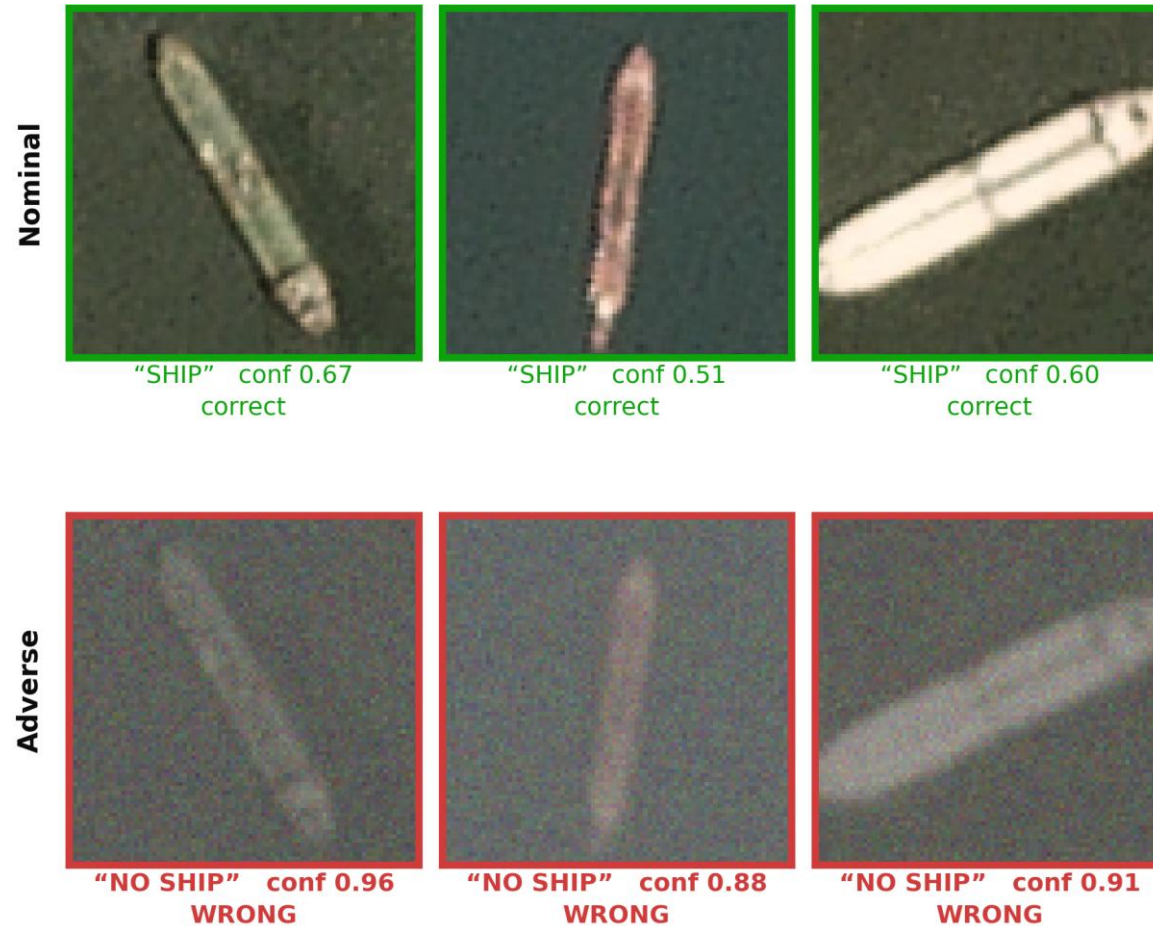


"SHIP" conf 0.51
correct



"SHIP" conf 0.60
correct

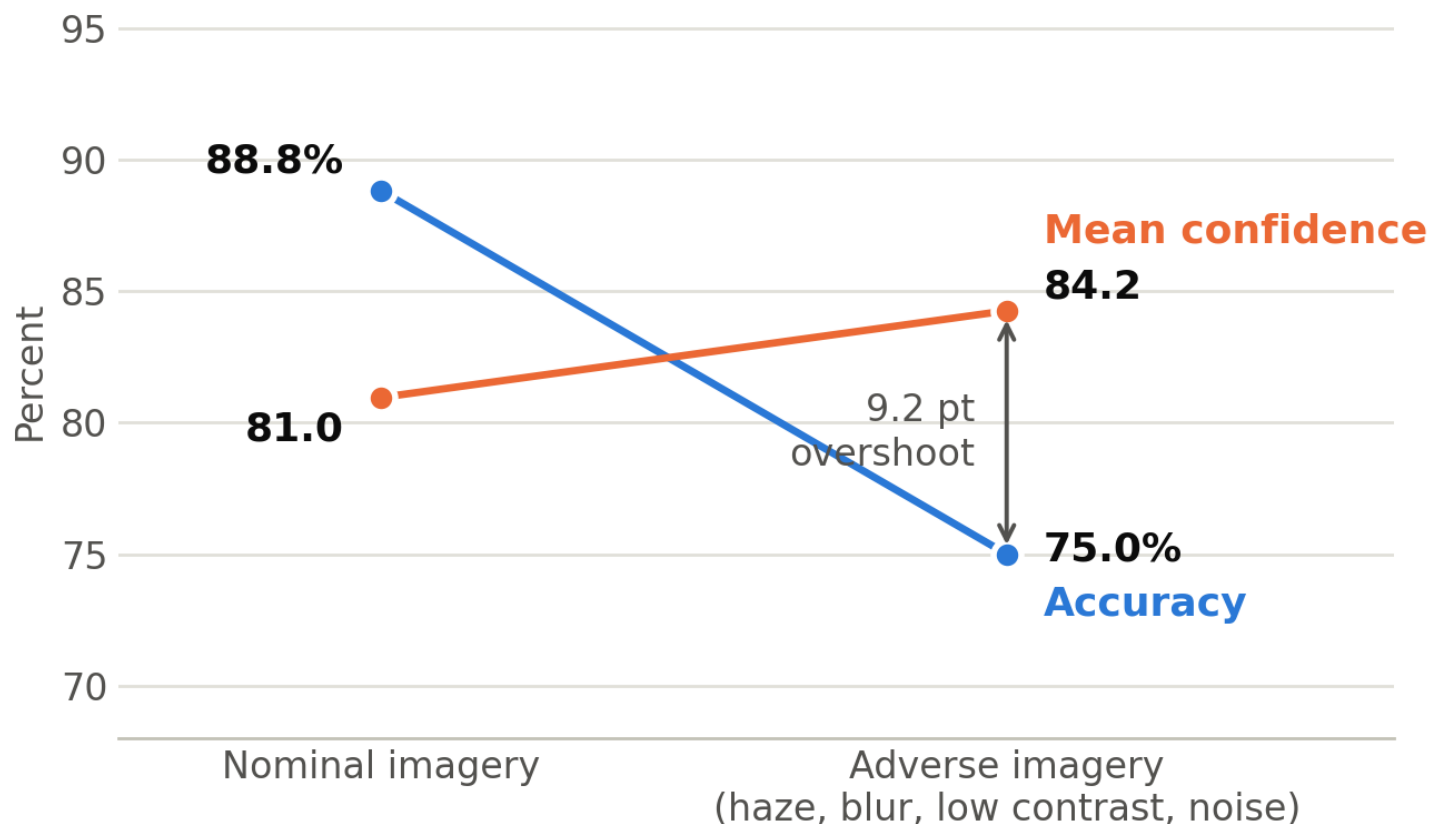
Add haze, and it stops seeing them.



Every chip still contains a ship. The model is now **wrong on all three, and far more confident.**

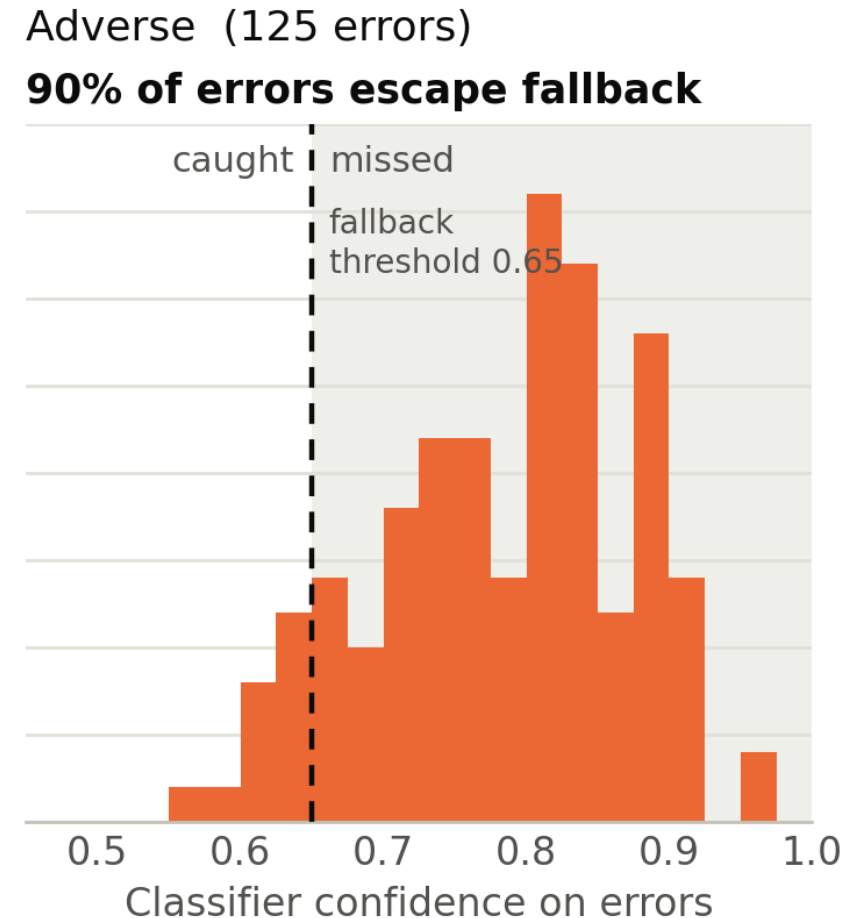
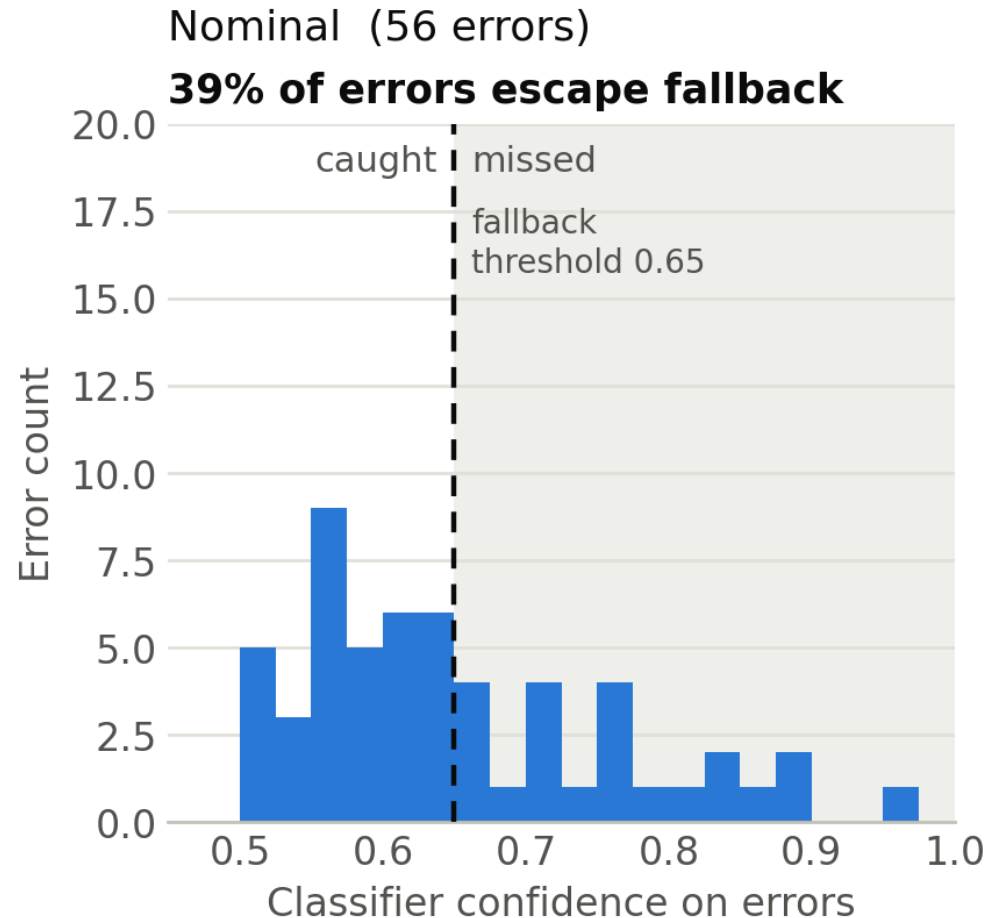
It is not three unlucky chips.

500 held-out images, nominal versus adverse.



Accuracy falls 13.8 points. Mean confidence **rises** 3.2 points.

So a confidence gate cannot see the errors.



Under adverse conditions **90% of errors sit above the 0.65 threshold**, invisible to it.

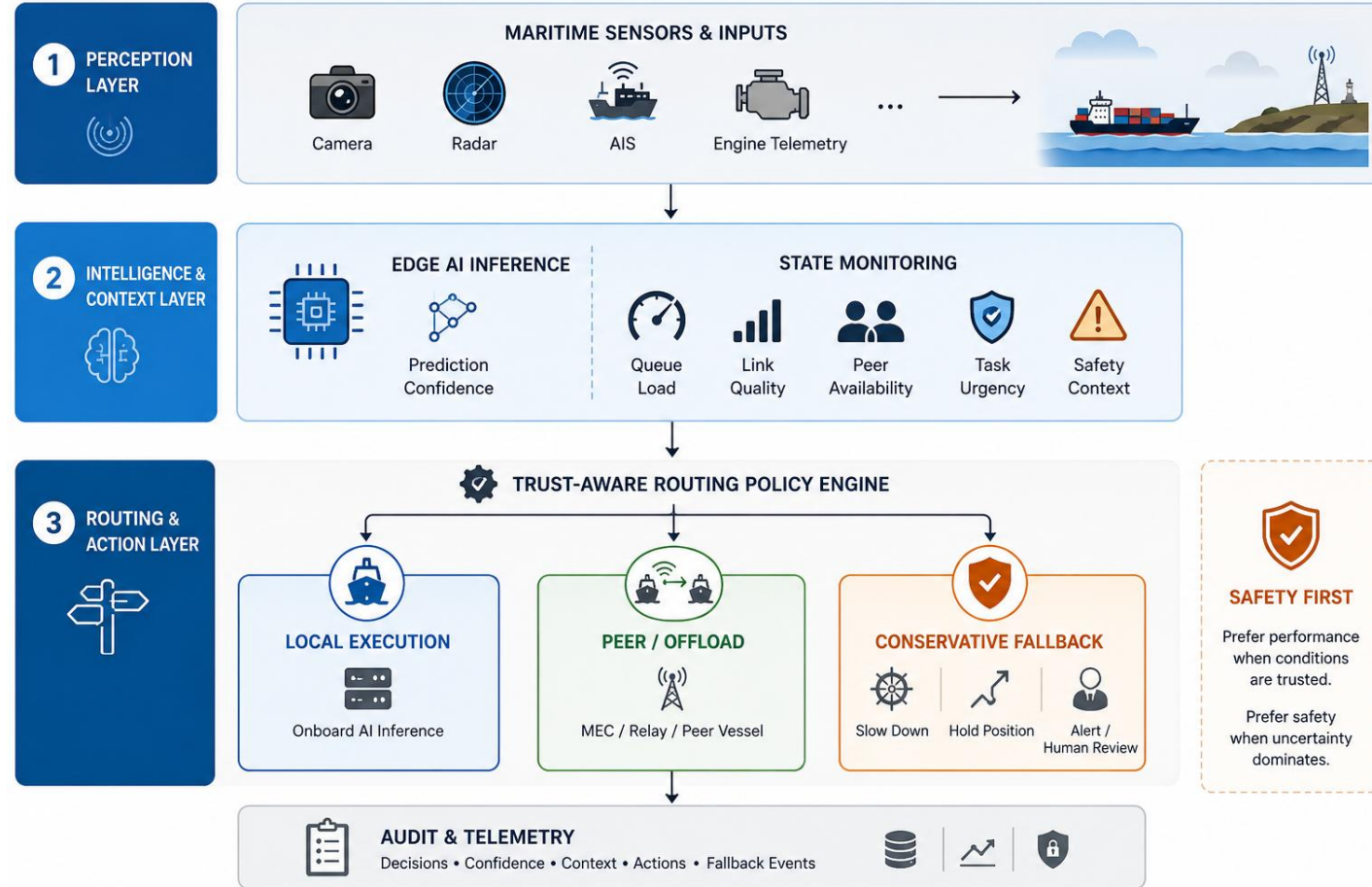
**A confidence gate fails
exactly when you need it.**

CONFROUTE

Make refusal a first-class routing action, alongside compute and offload.

One layer, between the model and the actuator.

Consumes uncertainty without touching the model. Logs every decision for audit.



Three actions, one of them new.

LOCAL

Run the model here.

Confident enough to act alone.

OFFLOAD

Ask a peer or relay.

Link is usable, queue is loaded.

FALLBACK

Refuse, and act safely.

Uncertain and cut off.

Scores decide, **guardrails override**: no offload without a peer, no local execution on a safety-critical task below the confidence floor.

Fallback is an action, not an error code.

What the conservative branch means, per maritime application.

Collision avoidance

Reduce speed, sound horn, hold heading, alert the operator

Maritime surveillance

Flag the chip for review, defer the alert, request another pass

Engine anomaly

Report “inspection needed”, fall back to a limit-based alert

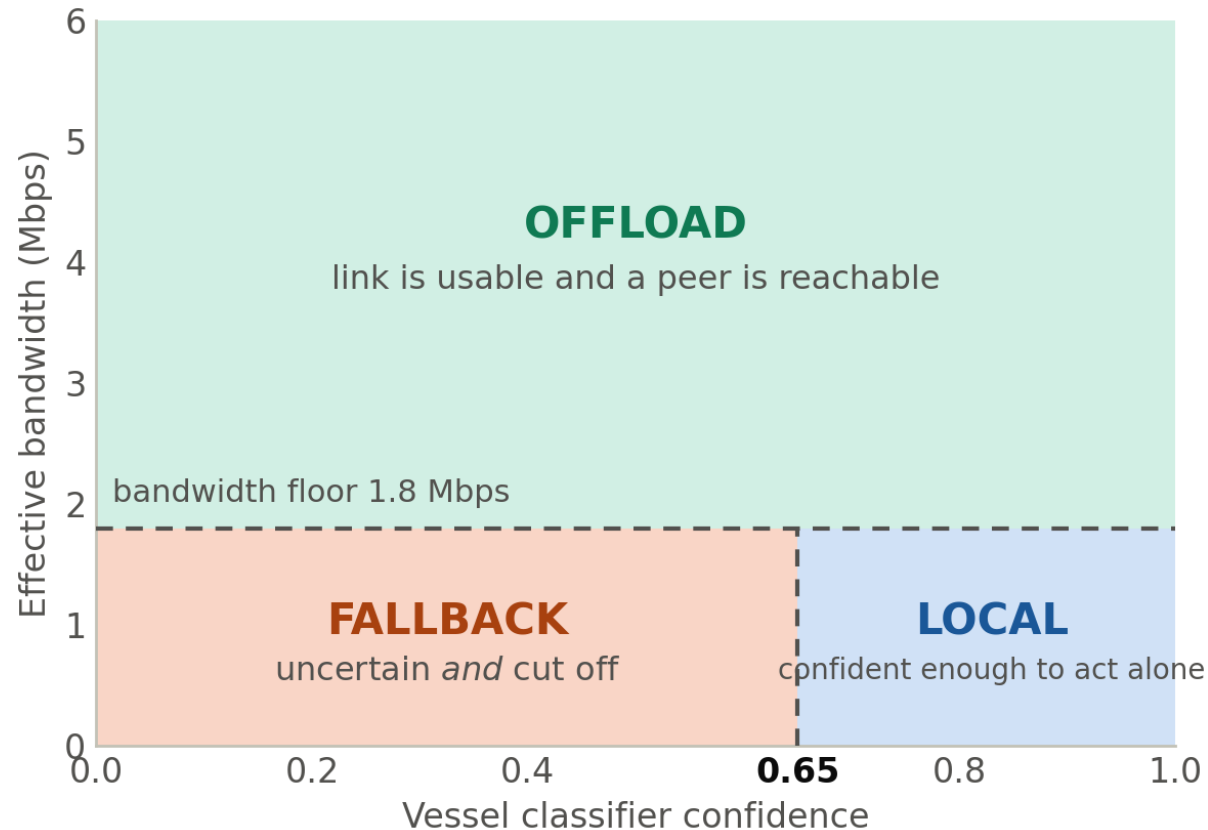
Waypoint decision

Hold position, revert to last safe heading, request human review

Conservative, auditable, and aligned with COLREGS/IMO practice: the ship does something safe and says why.

Refuse only when both signals fail.

Confidence below 0.65 and bandwidth below 1.8 Mbps.



If the link is healthy, the router would rather **ask a better-resourced peer than refuse**. That keeps fallback rare.

37,500 routing decisions.

2,500 tasks × 5 policies × 3 link regimes.

4,000

PlanetScope chips
80 × 80, ShipsNet

ResNet-18

ImageNet-initialised
84.8% validation

3

link regimes
stable / intermittent /
degraded

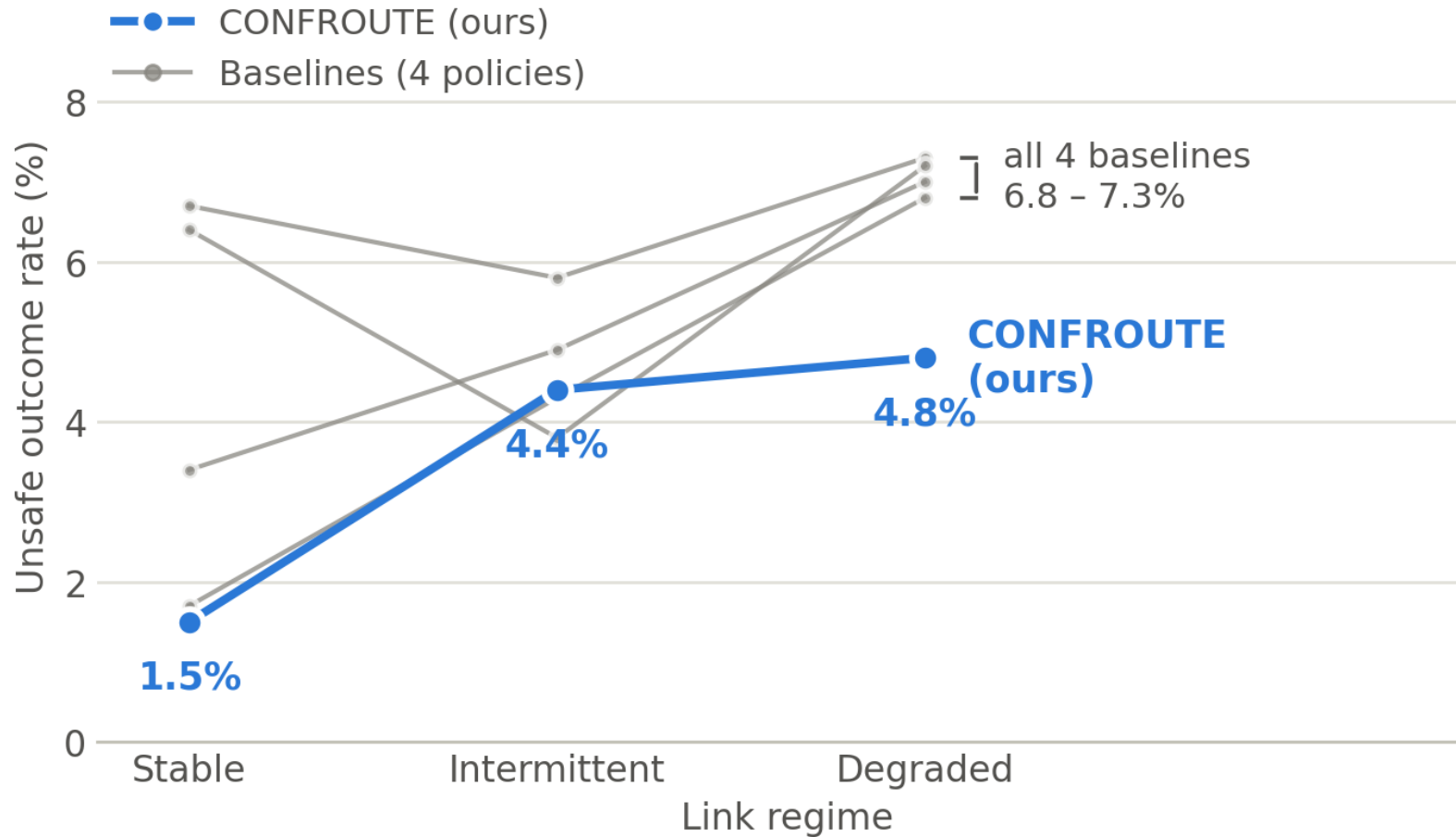
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baselines
local, offload, confidence,
load

Unsafe rate = routing-layer risk: safety-sensitive local errors, low-confidence local decisions, failed safety-sensitive offloads.

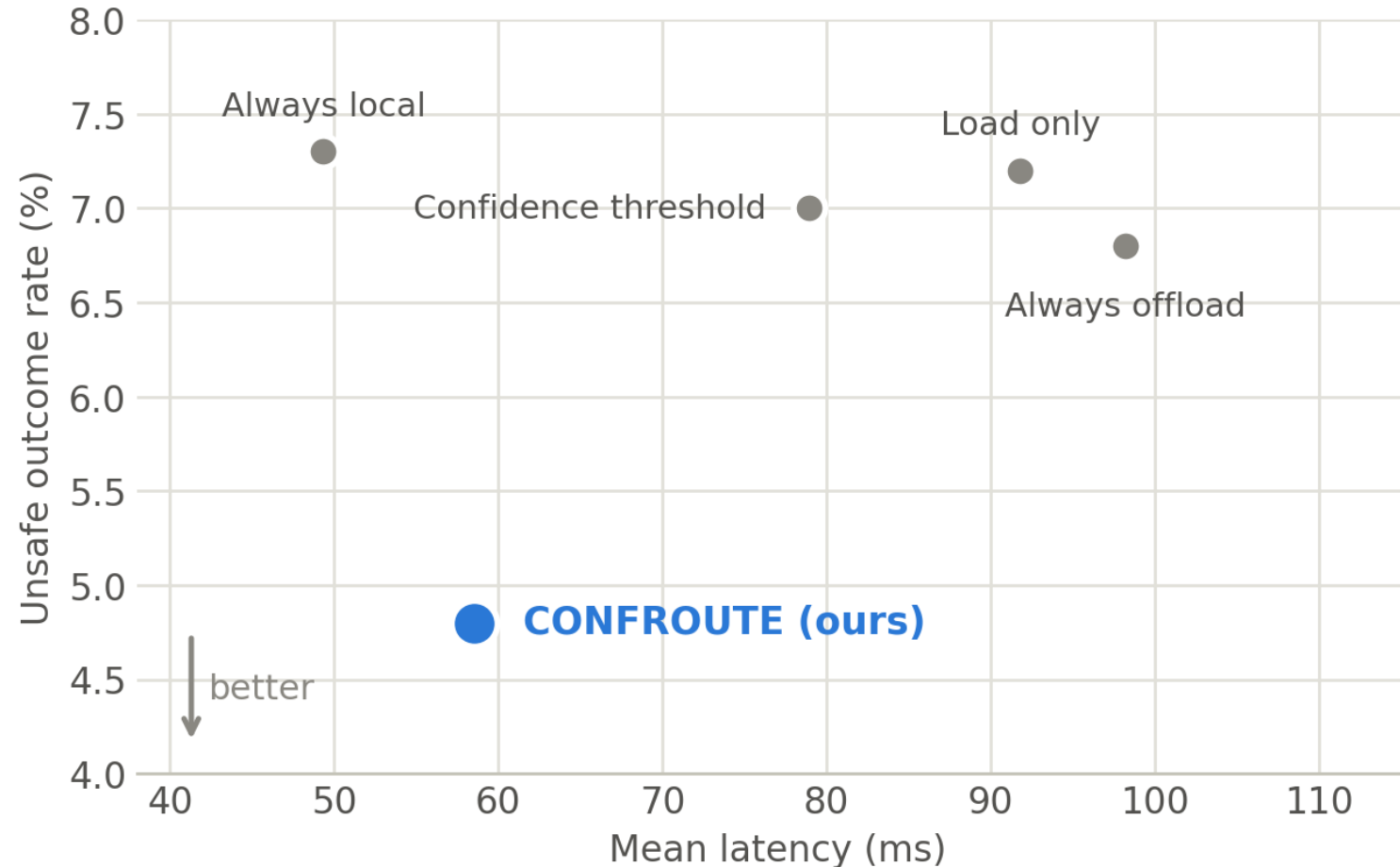
It is not a count of physical incidents.

Every baseline converges at about 7%.



When the link goes, so does their escape hatch. **CONFROUTE holds at 4.8%.**

Degraded links: safer and faster.



Unsafe 6.8% → 4.8% and latency 98.2 → 58.5 ms against always-offload. **A failed offload costs both.**

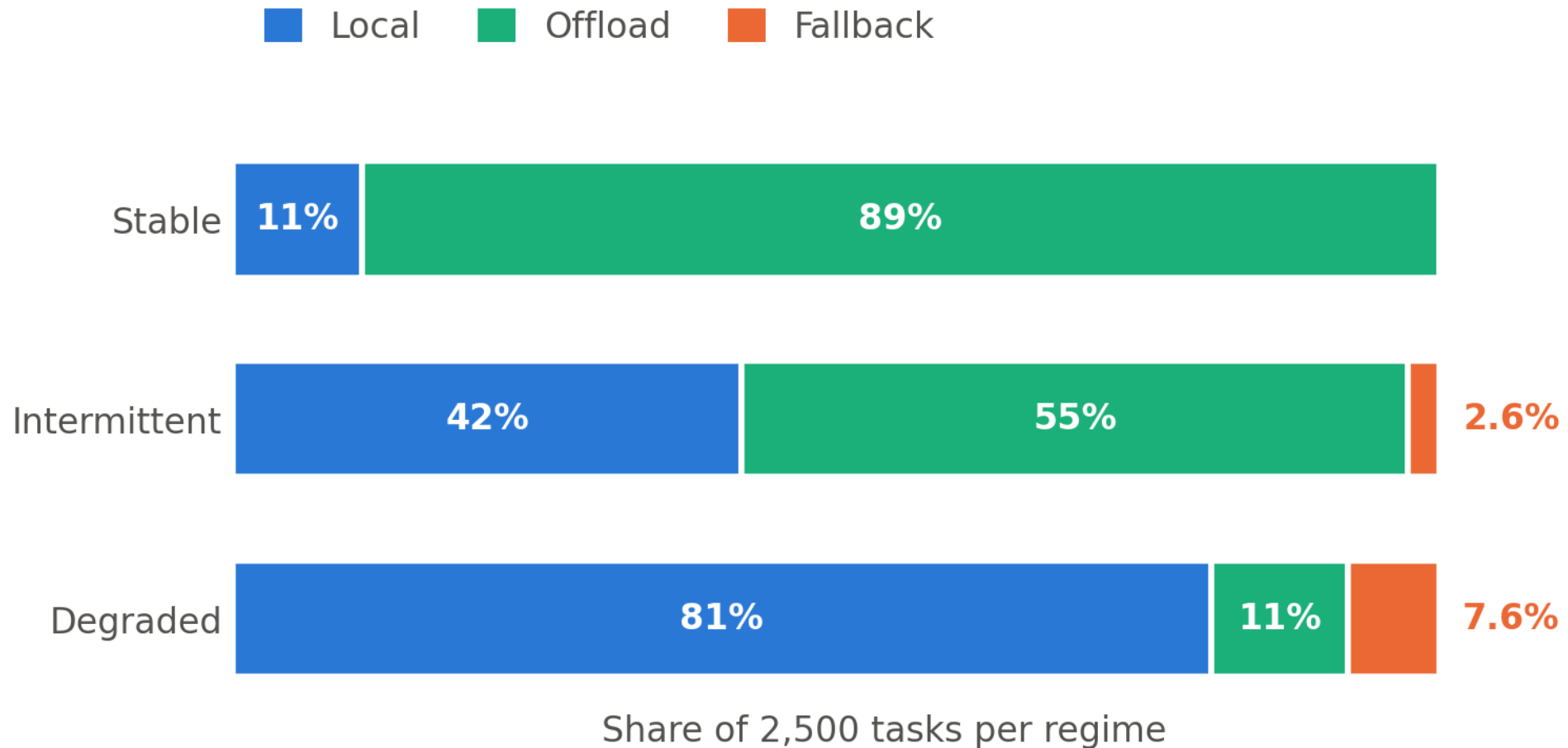
Where we do not win: intermittent links.

The regime in which offloading still mostly works.

Policy	Latency (ms)	Unsafe rate	Fallback
Always local	49.4	5.8%	0.0%
Always offload	71.6	4.3%	0.0%
Confidence threshold	59.6	4.9%	0.0%
Load only	67.0	3.8%	0.0%
CONFROUTE (ours)	61.6	4.4%	2.6%

Load-only routing is safer here (3.8% vs 4.4%), at 5.4 ms more latency. Our case is the degraded regime, which is also the regime that actually threatens the vessel.

The router shifts with the link.



Offload collapses 89% → 11%. Fallback stays a **bounded minority**: the vessel keeps operating.

Both parts are load-bearing.

Unsafe outcome rate under degraded links.

Full CONFROUTE	4.8%	
Without the fallback action	6.2%	p = 0.012
Without link-state awareness	5.9%	p = 0.037

The routing layer was never the bottleneck.

We improved it, and unsafe outcomes still occur at high confidence.

What we are not claiming.

Limitations

- Degradation is synthetic.
- One task family: satellite chips.
- Outcomes from a cost model.
- Thresholds fixed and trace-tuned.

Where this goes

- Calibration first: conformal, OOD.
- Adaptive thresholds via bandits.
- Peer-aware forwarding.
- Real traces, hardware-in-the-loop.

Three things to take away.

1 **Fallback deserves to be a routing action.**

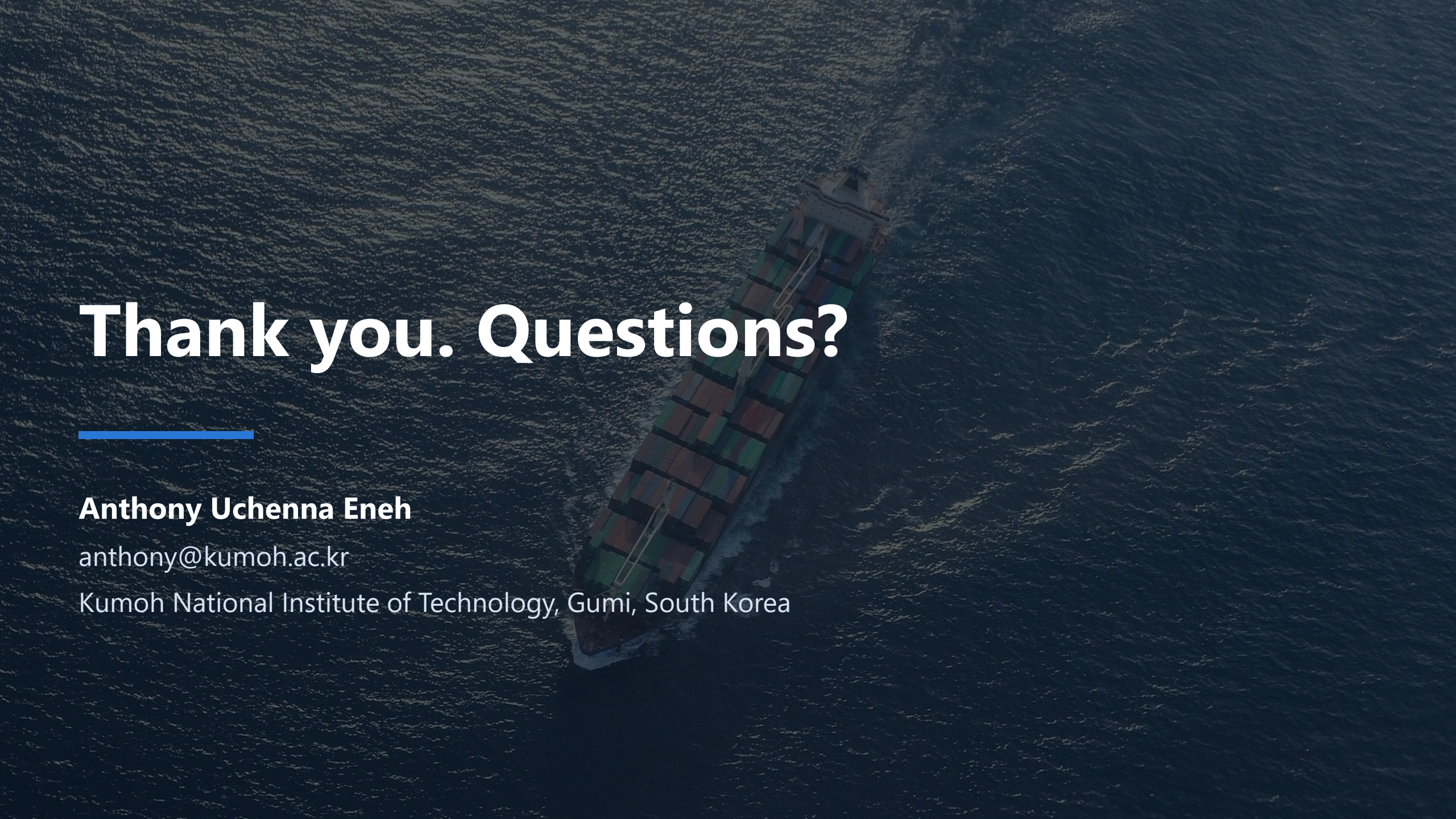
Under degraded links it cuts unsafe outcomes from 6.8% to 4.8% and saves 40 ms.

2 **Confidence thresholds break exactly when needed.**

Accuracy falls 13.8 points while confidence rises, and 90% of errors clear the gate.

3 **Calibration is the reliability bottleneck.**

Ahead of routing or scheduling. Safe autonomy at sea depends on knowing what the model does not know.

An aerial photograph of a large container ship sailing on the ocean. The ship is viewed from above, showing its deck covered with many colorful shipping containers in shades of red, green, and blue. The ship is moving towards the bottom left of the frame, leaving a white wake behind it. The ocean is a deep blue with visible ripples and small waves.

Thank you. Questions?

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Backup

Backup: full results

Policy	Regime	Latency (ms)	Unsafe	Energy (J)
Always local	Stable	49.3	6.7%	0.50
Always offload	Stable	33.4	1.7%	0.60
Confidence threshold	Stable	46.5	3.4%	0.53
Load only	Stable	48.2	6.4%	0.51
CONFROUTE	Stable	33.5	1.5%	0.60
Always local	Intermittent	49.4	5.8%	0.50
Always offload	Intermittent	71.6	4.3%	0.59
Confidence threshold	Intermittent	59.6	4.9%	0.53
Load only	Intermittent	67.0	3.8%	0.58
CONFROUTE	Intermittent	61.6	4.4%	0.56
Always local	Degraded	49.3	7.3%	0.50
Always offload	Degraded	98.2	6.8%	0.56
Confidence threshold	Degraded	78.9	7.0%	0.55
Load only	Degraded	91.8	7.2%	0.56
CONFROUTE	Degraded	58.5	4.8%	0.50

Backup: confidence traces

Nominal	500 samples	88.8% accuracy	mean confidence 0.810	std 0.140
Adverse	500 samples	75.0% accuracy	mean confidence 0.842	std 0.085

The adverse transform is deterministic: reduced contrast and brightness, Gaussian blur, haze blending, sensor-like noise.

Note the narrowing standard deviation: the model is not just wrong, it is uniformly confident.

Mean confidence on incorrect predictions: 0.651 nominal, 0.782 adverse.

Errors above the 0.65 threshold: 22 of 56 nominal, 113 of 125 adverse.

Image credits

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Aerial container ship (title, closing)

Wilfredor, CC BY-SA 3.0

Vessel in haze

Cavernia, CC BY-SA 4.0

Bulk carrier in heavy seas

Web107 / Georgfotoart, CC BY-SA 3.0

Shipborne radomes

High Contrast, CC BY 3.0 de

Frigate radar masts

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Satellite imagery and 80 × 80 chips: Ships in Satellite Imagery (ShipsNet), Planet Labs / Kaggle, CC BY-SA 4.0.